12TH BIOLOGY · CHAPTER 18 · SOLVED EXERCISE

Inheritance

Complete textbook exercise: MCQs, Short Questions, Long Questions and Inquisitive Questions

SECTION 1: MULTIPLE CHOICE QUESTIONS

1. In the monohybrid cross of Mendel, when a pure round-seeded plant was crossed with a winkled-seeded plant, all the F_1 progeny were round-seeded, because of:

(a) Co-dominance
(b) Incomplete dominance
(c) Complete dominance
(d) Epistasis

Answer: (c) Complete dominance — The dominant allele (R) is fully expressed and the recessive allele (r) is not expressed at all in the heterozygote (Rr).

2. In a cross of AaBb x AaBb, what fraction of the offspring can be expected to express one of the two dominant alleles, but not both?

(a) 1/16
(b) 6/16
(c) 9/1
(d) 16/16

Answer: (b) 6/16 — A_bb = 3/16 and aaB_ = 3/16, so the total is 3/16 + 3/16 = 6/16.

Option (c) is printed in the textbook as "9/1"; it is probably a misprint for 9/16.

3. In which of these cases, two genes can experience independent assortment?

(a) They are located in close proximity on the same chromosome.
(b) Crossing over between the genes does not occur.
(c) They are located on different chromosomes or are far apart on the same chromosome.
(d) The expression of one gene does not affect the expression of the other.

Answer: (c) They are located on different chromosomes or are far apart on the same chromosome. — Genes on different chromosomes, or far apart on the same chromosome, assort independently.

4. The blood group AB may be given to a person who has blood group;

(a) A
(b) O
(c) B
(d) AB

Answer: (d) AB — AB blood has both antigens, so it can be given only to an AB recipient, who has no anti-A or anti-B antibodies.

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5. If father has blood group B and the mother has blood group A, what blood groups their children have?

(a) A or B only
(b) AB only
(c) AB or O only
(d) A, B, AB or O

Answer: (d) A, B, AB or O — If the parents are heterozygous ($I^Bi \times I^Ai$), the children can be I^AI^B (AB), I^Ai (A), I^Bi (B) or ii (O).

6. A gene for corn has two alleles, one for yellow kernels and one for white kernels. Cross pollination of yellow corn and white corn results in ears with approximately equal number of yellow and white kernels. Which term best describes the relationship between the two alleles?

(a) Crossing over
(b) Recombination
(c) Incomplete dominance
(d) Codominance

Answer: (d) Codominance — Both alleles are fully expressed side by side (yellow and white kernels), and there is no blending into an intermediate colour.

The textbook does not discuss this corn example. (d) is chosen because both colours are fully expressed without blending.

7. A cross between two plants with genotypes AaBb and AaBb results in an F_1 generation that is 25% AABB, 50% AaBb, and 25% aabb. Which reason most likely explains why other possible genotypes are not present?

(a) The genes underwent independent assortment
(b) The loci of the genes are close together.
(c) The loci of the genes are on different chromosomes.
(d) Crossing over occurred between chromosomes.

Answer: (b) The loci of the genes are close together. — Only parental combinations (AB and ab) appear, which shows complete linkage: the genes lie very close on the same chromosome.

8. In pea plants, the allele for tall stalks (T) is dominant over the allele for short stalks (t). Suppose a cross between a tall pea plant and a short pea plant produces 43 tall offspring and 47 short offspring. If one of the tall offspring is crossed with one of the short offspring, what ratio of genotypes would be most likely in their offspring?

(a) 1 TT : 2 Tt : 1 tt
(b) 3 Tt : 1 tt
(c) All Tt
(d) 1 Tt : 1 tt

Answer: (d) 1 Tt : 1 tt — The 1:1 result shows the tall parent was Tt, so every tall offspring is Tt. Tt × tt gives 1 Tt : 1 tt.

9. Allele B produces black coat colour while b produces white coat colour. Allele C produces curly fur while c produces straight fur. In a cross between two BbCc parents, predict the fraction of offspring with black coat colour and straight fur.

(a) 1/16 (b) 9/16
(c) 3/4 (d) 16/16

Answer: None of the printed options is correct. The correct fraction is 3/16.

Black coat (B_) = 3/4 and straight fur (cc) = 1/4. By the product rule, black and straight = 3/4 × 1/4 = **3/16**.

This is a textbook error: 3/16 is missing from the options (1/16, 9/16, 3/4, 16/16). 9/16 is the fraction of black and curly offspring.

10. What is the criterion to mark an allele as dominant?

(a) The frequency of the allele in population
(b) The expression of the allele in heterozygous genotype
(c) The location of the allele on chromosome
(d) The frequency of the crossing over of the allele

Answer: (b) The expression of the allele in heterozygous genotype — A dominant allele is the one that is expressed in the heterozygous individual.

11. Which of these characters is controlled by a multiple allelic system?

(a) ABO blood group system (b) Rh blood group system
(c) MNS blood group system (d) All of these

Answer: (a) ABO blood group system — The ABO system has three alleles (I^A, I^B, i). In the textbook, the Rh system (D, d) and the MN system have two alleles each.

12. Albinism is caused by a recessive gene. A woman with albino father marries an albino man. What will be the proportion of her progeny?

(a) 2 normal : 1 albino (b) All normal
(c) All albino (d) 1 normal : 1 albino

Answer: (d) 1 normal : 1 albino — The normal woman got allele a from her albino father, so she is Aa. Aa × aa gives 1 Aa (normal) : 1 aa (albino).

13. When the activity of one gene is suppressed by the activity of a non-allelic gene, it is known as;

(a) Incomplete dominance (b) Epistasis
(c) Complete dominance (d) Multiple allelic effect

Answer: (b) Epistasis — In epistasis, an allele masks or modifies the effect of the alleles of another gene at another locus.

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14. Which of the following is a result of Erythroblastosis fetalis?

(a) Anaemia in the foetus (b) Jaundice in the foetus
(c) Mental retardation in the foetus (d) All of these

Answer: (d) All of these — Destruction of foetal RBCs causes haemolytic anaemia; bilirubin causes jaundice and can damage brain cells.

15. Colour blindness is more common in males because it is a/an;

(a) Autosomal trait (b) X-linked dominant trait
(c) X-linked recessive trait (d) Y-linked trait

Answer: (c) X-linked recessive trait — A male has only one X chromosome, so a single recessive allele is enough to express the trait.

SECTION 2: SHORT QUESTIONS

1. Name the main exceptions to the Mendel's laws of inheritance.

- Incomplete dominance
- Co-dominance
- Multiple alleles
- Polygenic inheritance
- Epistasis
- Gene linkage and crossing over
- Pleiotropy
- Sex linkage

2. What is probability? How does probability relate to genetics?

Probability means how often a particular outcome occurs. It is calculated by dividing the number of times an event occurs by the total number of possible outcomes.

The transfer of genes from one generation to the other is random. So the possible new allelic combinations in the offspring are calculated by mathematical probability, using a Punnett square or the product and sum rules. For example, in Aa × Aa the probability of an aa offspring is 1/2 × 1/2 = 1/4.

3. Justify that multiple alleles provide many different phenotypes.

When a gene has only two alleles, only three genotypes are possible. When a gene has more than two alleles, more combinations of alleles are possible, so more phenotypes appear. For example, the ABO gene has three alleles (I^A, I^B and i). They make six genotypes (I^AI^A, I^Ai, I^BI^B, I^Bi, I^AI^B, ii) and four phenotypes (blood groups A, B, AB and O).

4. Name various human blood group systems.

The International Society of Blood Transfusion recognizes 43 blood group systems in humans. The important ones are: ABO system, Rh system, MNS system, Lutheran system, Kell system, Lewis system and Duffy system.

5. What do you mean by multiple alleles? Name the alleles responsible for ABO blood group system.

When a gene has more than two alternative forms (alleles) in a population, they are called multiple alleles. The ABO blood group system is controlled by gene I, which has three alleles: $\mathbf{I^A}$, $\mathbf{I^B}$ and **i**.

6. What is co-dominance. Give an example.

Co-dominance is the type of inheritance in which both alleles of a gene pair are fully expressed in a heterozygous individual, without blending or masking. Example: in the human MN blood group, a person with genotype L^ML^N has both M and N antigens on the RBCs (MN blood group). Blood group AB (I^AI^B) is another example.

7. What antigens and antibodies are found in ABO blood group system?

Blood group	Genotypes	Antigen on RBCs	Antibodies in plasma
A	I^AI^A or I^Ai	Antigen-A	Anti-B
B	I^BI^B or I^Bi	Antigen-B	Anti-A
AB	I^AI^B	Antigen-A and antigen-B	None
O	ii	None	Anti-A and anti-B

8. Justify why O-negative individuals are called universal donors and AB-positive individuals are called universal recipients.

O-negative is the actual universal donor because its RBCs have neither any antigen of the ABO system (A or B) nor the antigen of the Rh system. So no antibody of the recipient can attack the donated RBCs.

AB-positive is the actual universal recipient because its plasma has neither anti-A and anti-B nor anti-Rh antibodies. So it can receive any donor's blood.

9. Why is human male referred as heterogametic?

A human male has XY sex chromosomes. So he produces two types of sperm: half carry an X chromosome and half carry a Y chromosome. Because he forms two different types of gametes, he is called heterogametic.

10. What is gene linkage?

Genes present on the same chromosome are called linked genes, and this phenomenon is known as gene linkage. Closely linked genes tend to stay together and do not assort independently during gamete formation.

11. How can the linked gene assort independently?

Linked genes can be separated by **crossing over**. During meiosis-I, the chromatids of homologous chromosomes exchange sections of DNA, so the alleles of linked genes are rearranged (recombination). If the linked genes are far apart on the same chromosome, crossing over between them is so frequent that they assort independently.

12. List at least five polygenic traits in humans.

1. Height
2. Skin colour
3. Weight
4. Intelligence
5. Eye colour

The textbook names height, skin colour, weight and intelligence. Eye colour is added from general knowledge to complete five.

13. Write brief note on Y-linked inheritance in humans.

The Y chromosome contains relatively few genes compared with the X chromosome, so Y-linked traits are relatively rare. Y-linked genes do not have alleles on the X chromosome. Such genes are called **holandric genes** (or hemizygous genes). They are located only on the non-homologous region of the Y chromosome. Y-linked traits appear only in males and are inherited from father to son only. Examples in humans: hypertrichosis (growth of hair on the rim of the pinna), porcupine man (straight hair on the body) and webbing of toes.

14. Define sex linkage and give an example.

The phenomenon in which traits are controlled by alleles present on the sex chromosomes is called sex linkage. Such traits are called sex-linked traits. Examples: white eye colour in *Drosophila*; haemophilia and red-green colour blindness in humans.

15. Solve the following genetic problems:

i. A plant of genotype AABbCC is self-pollinated. Calculate the phenotypic ratio of F_2 generation.

The plant is homozygous for A and C, so all offspring are AA and CC. Only gene B segregates: Bb × Bb.

Gametes	ABC	AbC
ABC	AABBCC	AABbCC
AbC	AABbCC	AAbbCC

Phenotypic ratio = 3 (dominant for B) : 1 (recessive for B). All offspring show the dominant traits of A and C.

ii. In four-o'clock plant, the red and white flower colours show incomplete dominance. Calculate the ratios in crosses between (i) two red-flowered plants; (ii) a red-flowered and a pink-flowered plant; (iii) a pink-flowered and a white-flowered.

Cross	Offspring genotypes	Ratio
(i) Red × Red (RR × RR)	All RR	All red
(ii) Red × Pink (RR × RW)	1 RR : 1 RW	1 red : 1 pink
(iii) Pink × White (RW × WW)	1 RW : 1 WW	1 pink : 1 white

iii. A man with blood group A marries a woman who has blood group AB. What blood groups are expected among their children?

The woman is I^AI^B. The man may be I^AI^A or I^Ai.

Case 1 ($I^AI^A \times I^AI^B$): children are I^AI^A (A) and I^AI^B (AB) in a 1:1 ratio.

Case 2 ($I^Ai \times I^AI^B$):

Gametes	I^A	I^B
I^A	I^AI^A (A)	I^AI^B (AB)
i	I^Ai (A)	I^Bi (B)

Result: The children can have blood groups A, AB or B (2 A : 1 AB : 1 B in case 2). Blood group O is not possible.

iv. Two white sheep produce a black offspring. What are the parents' genotypes for the colour? What is the probability that their next offspring will be black?

Black appeared from two white parents, so black is recessive (w) and white is dominant (W). The black offspring is ww and got one w from each parent. So both parents are heterozygous: **Ww × Ww**.

Offspring: 1 WW : 2 Ww : 1 ww. Probability that the next offspring is black (ww) = 1/2 × 1/2 = **1/4 (25%)**.

v. Albinism (lack of pigment) in man is caused by a recessive gene. If normal parents have an albino child, what is the probability that their next child will be normal for colour?

The albino child is aa, so both normal parents are carriers: Aa × Aa. Offspring: 1 AA : 2 Aa : 1 aa. Probability of a normal child (AA or Aa) = 1/4 + 1/4 + 1/4 = **3/4 (75%)**.

16. Differentiate between:

• Gene and allele

Gene	Allele
A sequence of nucleotides, as part of DNA, which codes for the formation of a polypeptide.	An alternative form of a gene, leading to an alternative trait.
It controls a character, e.g., seed shape.	It controls one form of the character, e.g., round (R) or wrinkled (r).

• Genotype and phenotype

Genotype	Phenotype
The combination of alleles in an individual.	The observable character or trait.
It is the blueprint.	It is the visible outcome.
Example: RR, Rr or rr.	Example: round seed or wrinkled seed.

• Homozygous and heterozygous

Homozygous	Heterozygous
The individual has a gene pair with similar alleles.	The individual has a gene pair with different alleles.
It is formed when gametes with the same alleles fuse.	It is formed when gametes with different alleles fuse.
Example: RR or rr.	Example: Rr.

• Monohybrid and dihybrid cross

Monohybrid cross	Dihybrid cross
Inheritance of one character is studied.	Inheritance of two characters is studied together.
Example: round × wrinkled seeds (RR × rr).	Example: round yellow × wrinkled green (RRYY × rryy).
F_2 phenotypic ratio is 3:1.	F_2 phenotypic ratio is 9:3:3:1.
It led to the law of segregation.	It led to the law of independent assortment.

• Incomplete dominance and co-dominance

Incomplete dominance	Co-dominance
In the heterozygous state, both genes blend their phenotypic effects.	In the heterozygous state, both genes fully express their effects.
The heterozygote shows an intermediate phenotype between the two parental phenotypes.	The heterozygote shows both parental phenotypes at a time.
Example: flower colour in Japanese 4 O'clock plant.	Example: human MN blood group and human AB blood group.

• Epistasis and polygenic inheritance

Epistasis	Polygenic inheritance
One allele masks or modifies the effect of another gene.	Many genes act together, each having a small additive effect, to produce a trait.
It usually produces discontinuous variation (distinct categories).	It results in continuous variation.
Example: coat colour in Labrador (9:3:4).	Example: grain colour in wheat; skin colour in man.

• XX-XY and XX-ZO sex determination

XX-XY system	XX-XO system
Male has X and Y chromosomes.	Male has only one X chromosome; there is no Y chromosome.
Sperms are of two types: X-carrying and Y-carrying.	Sperms are of two types: X-carrying and nullo (without a sex chromosome).
Male and female have the same number of chromosomes.	Male has one chromosome less than the female (grasshopper: male 23, female 24).
Found in *Drosophila*, mammals and most other vertebrates.	Found in some insects, e.g., grasshopper.

"XX-ZO" is printed in the textbook. The chapter describes the XX-XO system, so it is answered as XX-XO.

• Sex-linked and sex-limited traits

Sex-linked traits	Sex-limited traits
Their alleles are present on the sex chromosomes.	Their alleles are not present on the sex chromosomes.
They can appear in both sexes, but in different ratios.	They are expressed in only one sex, though the genes are present in both.
Examples: haemophilia, colour blindness.	Examples: beard in men, milk production in female mammals.

• Sex-limited and sex-influenced traits

Sex-limited traits	Sex-influenced traits
The genes are expressed in only one sex.	The same gene is expressed in both sexes, but differently.
Due to anatomical or hormonal differences.	The allele is dominant in one sex and recessive in the other, due to hormonal difference.
Examples: horns in male sheep, beard in men.	Examples: baldness and amount of body hair in humans.

• XX-XY and ZZ-ZW sex determination

XX-XY system	ZZ-ZW system
Female is XX and male is XY.	Female is ZW and male is ZZ.
Male is heterogametic (two types of sperm).	Female is heterogametic (two types of eggs).
The sperm determines the sex of the offspring.	The egg determines the sex of the offspring.
Found in *Drosophila* and mammals.	Common in birds.

SECTION 3: LONG QUESTIONS

1. Explain Mendel's experiment which led to the formulation of the law independent assortment.

Mendel studied two characters together: seed shape, round (R) dominant and wrinkled (r) recessive; and seed colour, yellow (Y) dominant and green (y) recessive.

1. He crossed pure-breeding round yellow plants (RRYY) with pure-breeding wrinkled green plants (rryy).
2. All F_1 seeds were round and yellow, with the genotype RrYy. These F_1 plants are called **dihybrids**.
3. He self-fertilized the F_1 plants. Each dihybrid produced four kinds of gametes: RY, Ry, rY and ry.

4. When 4 male × 4 female gametes combined, 16 combinations were formed in F_2.

F_1 gametes	RY	Ry	rY	ry
RY	RRYY	RRYy	RrYY	RrYy
Ry	RRYy	RRyy	RrYy	Rryy
rY	RrYY	RrYy	rrYY	rrYy
ry	RrYy	Rryy	rrYy	rryy

F_2 phenotypic ratio: 9 round yellow : 3 round green : 3 wrinkled yellow : 1 wrinkled green.

The results showed that the genes for seed shape assort independently of the genes for seed colour. So Mendel proposed the **law of independent assortment**: “each pair of alleles assorts independently of other pairs of alleles during gamete formation”.

2. Explain incomplete dominance and exemplify it through the inheritance of flower colour in 4 O’ clock plant.

In incomplete dominance, neither allele is fully dominant over the other. The heterozygous individual shows a phenotype that is **intermediate** between the phenotypes of both parents.

Example: In Japanese four-o’clock plants, when a red-flowered plant (RR) is crossed with a white-flowered plant (WW), all the F_1 offspring are heterozygous and pink-flowered (RW). This shows that neither the allele for red (R) nor that for white (W) is completely dominant. When two pink-flowered plants (RW × RW) are crossed, the F_2 shows red (RR), pink (RW) and white (WW) flowers in a 1:2:1 ratio.

F_1 gametes (RW × RW)	R	W
R	RR (red)	RW (pink)
W	RW (pink)	WW (white)

3. Explain Erythroblastosis foetalis in the light of antigen-antibody reaction. What measures can be taken to prevent Erythroblastosis foetalis?

Erythroblastosis foetalis is a haemolytic disease of the foetus or new-born caused by Rh incompatibility between an **Rh-negative mother** and her **Rh-positive foetus**.

Antigen–antibody reaction

1. An Rh-negative mother (dd) carries an Rh-positive baby (DD or Dd).
2. Foetal blood may enter the mother’s blood. The foetal RBCs carry the Rh-antigen.
3. On exposure to these Rh-antigens, the mother produces anti-Rh antibodies.
4. These antibodies can cross the placenta and enter the foetal blood.
5. There, the mother’s anti-Rh antibodies destroy the foetal RBCs. This leads to haemolytic anaemia and other serious complications.

Complications

In severe cases, the destruction of foetal RBCs can cause miscarriage or stillbirth. If the foetus survives, the liver and spleen become enlarged because they try to make more RBCs. The breakdown of RBCs produces bilirubin, which builds up and causes jaundice. High bilirubin can also damage brain cells.

Preventive measures

- When there are chances of Rh incompatibility, the mother is given an injection of **Rh-antiserum** (serum containing anti-Rh antibodies) during early pregnancy and immediately after birth. It destroys any of the baby's RBCs that have crossed into the mother's blood before they sensitize her immune system.
- The surviving baby's blood is replaced by Rh-negative blood that is free of anti-Rh antibodies.

4. Describe polygenic inheritance, using an example from grain colour in wheat.

Polygenic characters are controlled by many gene pairs, not just one. In polygenic inheritance, all the alleles of many genes contribute to the same trait in an **additive** way, and their combined effect produces a single trait. Polygenic traits show a range of differences, i.e., **continuous variation**, in a population.

Grain colour in wheat

Wheat grain colour shows continuous variation, from white to dark red. It is controlled by three polygenes: A, B and C. Each gene has a dominant allele (A, B, C) that produces red pigment and a recessive allele (a, b, c) that does not.

1. The genotype AABBCC produces the maximum red pigment, so the grains are dark red. The genotype aabbcc produces no pigment, so the grains are white.
2. When dark red wheat (AABBCC) is crossed with white wheat (aabbcc), the F_1 generation (AaBbCc) shows an intermediate red colour.
3. When F_1 plants are crossed, a wide range of colours appears in F_2. Out of 64, only one plant has the darkest red grains (AABBCC) and one has white grains (aabbcc). The other 62 plants show different shades of red.

Number of pigment-producing alleles	6	5	4	3	2	1	0
F_2 plants (out of 64)	1	6	15	20	15	6	1

Environmental factors like light, water and nutrients also influence the grain colour.

5. Explain epistasis with the help of an example.

Epistasis is the gene interaction in which an allele masks or modifies the effect of the alleles of other genes (present at other loci). It is a non-allelic interaction. The allele which masks or modifies the effect is called **epistatic**, and the allele whose effect is masked or modified is called **hypostatic**. For example, in humans, the allele of baldness is epistatic to the alleles of black and blonde hair.

Example: coat colour in Labrador

In Labrador dogs, coat colour (black, brown, yellow) depends on two genes.

1. **Gene B** controls pigment colour. Dominant B gives black, recessive b gives brown. So BB or Bb is black and bb is brown.
2. **Gene E** controls the deposition of pigment in fur. Dominant E allows pigment, but recessive e is epistatic to B or b: it blocks pigment deposition. So ee dogs have a yellow coat, whether B or b is present.

When two heterozygous black dogs (BbEe × BbEe) are crossed, the offspring are:

Genotypes	Fraction	Coat colour
B_E_	9/16	Black
bbE_	3/16	Brown
B_ee and bbee	3/16 + 1/16 = 4/16	Yellow

So the Mendelian 9:3:3:1 ratio is modified to **9 black : 3 brown : 4 yellow**.

6. Explain how gene linkage counters independent assortment and crossing-over modifies the progeny.

Gene linkage counters independent assortment

In body cells, thousands of genes are present on a limited number of chromosomes. So many genes are arranged in a line on each chromosome. Genes present on the same chromosome are called **linked genes**, and this is known as **gene linkage**. Closely linked genes tend to stay together and do not assort independently during gamete formation. Genes on different chromosomes, or those far apart on the same chromosome, assort independently.

According to independent assortment, a dihybrid forms four kinds of gametes in equal numbers and gives a 9:3:3:1 ratio in F_2. But linked genes travel together on one chromosome into the same gamete. So mostly (or only) parental combinations appear, and the 9:3:3:1 ratio is not obtained. In this way, linkage counters independent assortment.

Crossing over modifies the progeny

During the formation of gametes (meiosis-I), homologous chromosomes go through **crossing over**. The chromatids of two homologous chromosomes line up next to one another and exchange sections of DNA; the DNA strands actually break and re-join. After crossing over, the homologous chromosomes separate. The chromatids still have the same genes in the same order, but the alleles have been rearranged. When these recombinant chromatids are distributed to different gametes, a wide variety of gametes is produced. So crossing over leads to the recombination of linked genes and is responsible for genetic variability.

Because of crossing over, some gametes carry new (recombinant) combinations of linked alleles. So besides the parental types, a smaller number of new combinations appear in the progeny (incomplete linkage).

7. Exemplify the concept of gene linkage by quoting the example of wing length and width of abdomen in *Drosophila melanogaster*.

Thomas Hunt Morgan studied two traits in *Drosophila*: wing length and abdomen width. The allele for long wings (L) is dominant over vestigial wings (l), and the allele for broad abdomen (B) is dominant over narrow abdomen (b).

1. He crossed a fly with long wings and broad abdomen with a fly with vestigial wings and narrow abdomen. All F_1 flies had long wings and broad abdomens.
2. He then crossed two F_1 flies.
3. **Expected result:** According to the law of independent assortment, F_2 should show a 9:3:3:1 ratio: 9 long broad and 1 vestigial narrow (parental combinations), 3 long narrow and 3 vestigial broad (new combinations).
4. **Observed result:** Morgan observed only parental combinations in F_2: 3/4 flies with long wings and broad abdomen, and 1/4 with vestigial wings and narrow abdomen.

Conclusion: The genes for wing length and abdomen width are located on the same chromosome (linked), so they did not assort independently.

8. Explain the XX-XY mechanism of sex determination in *Drosophila* and mammals.

The sex of an animal is determined during its embryonic development, in most animals on the basis of the **karyotype** (the numbers, sizes and shapes of the complete set of chromosomes). The chromosomes which are similar in female and male are called **autosomes**. One pair is different in female and male and determines the gender; these are the **sex chromosomes**.

In *Drosophila*, mammals and most other vertebrates, the sex chromosome pair consists of a rod-shaped X chromosome and a hook-shaped Y chromosome. Females are XX and males are XY. *Drosophila* has 3 pairs of autosomes and 1 pair of sex chromosomes; humans have 22 pairs of autosomes and 1 pair of sex chromosomes.

Females are **homogametic**: all their eggs carry one X chromosome. Males are **heterogametic**: they produce two types of sperm, half with an X chromosome and half with a Y chromosome. If an X-carrying sperm fertilizes the egg, the zygote is XX and the baby is a female. If a Y-carrying sperm fertilizes the egg, the zygote is XY and the baby is a male.

Gametes	X (sperm)	Y (sperm)
X (egg)	XX — female	XY — male

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9. Describe the XX-XO and ZZ-ZW sex determination systems and evaluate by studying the karyotype.

XX–XO system

This system is found in some insects. For example, the grasshopper has 22 autosomes (11 pairs). The **male** has 23 chromosomes in total because its sex chromosome pair is XO, i.e., only an X chromosome is present and there is no Y chromosome. The male forms two types of sperm: half with X and half without any sex chromosome (**nullo gamete**). The **female** has 24 chromosomes; her sex chromosome pair is XX and all eggs carry one X. If an X-carrying sperm fertilizes the egg, an XX female is produced. If a nullo sperm fertilizes the egg, an XO male is produced.

ZW–ZZ system

This system is common in birds. The sex chromosomes are **ZW in females** and **ZZ in males**. The female bird produces two kinds of eggs, half with Z and half with W, while all sperms of the male have Z. When a Z-carrying egg is fertilized, the offspring is male (ZZ). When a W-carrying egg is fertilized, the offspring is female (ZW).

Evaluation from the karyotype

System	Karyotype of female	Karyotype of male	Heterogametic sex
XX–XO (grasshopper)	22 autosomes + XX = 24	22 autosomes + X = 23	Male (X and nullo sperms)
ZW–ZZ (birds)	Autosomes + ZW	Autosomes + ZZ	Female (Z and W eggs)

So, in a grasshopper karyotype, an individual with one chromosome less (a single X) is a male. In a bird karyotype, an individual with two different sex chromosomes (ZW) is a female.

The textbook sentence reads "When a Y carrying egg is fertilized by the sperm, offspring is female". It should be "W carrying egg".

10. Explain the inheritance of a sex-linked trait in *Drosophila*.

1. In 1910, T. H. Morgan observed a mutant male *Drosophila* with **white eyes** instead of the normal red. He crossed it with a normal red-eyed female. All F_1 flies had red eyes, so red is dominant over white.
2. He crossed the F_1 red-eyed flies with each other. Out of 4252 F_2 flies, 782 (18%) were white-eyed, and **all white-eyed flies were males**.
3. To test whether a white-eyed female could exist, he test-crossed an F_1 red-eyed female with the white-eyed male. He obtained red-eyed and white-eyed males and females in a 1:1:1:1 ratio. So a female can have white eyes.
4. He mated a white-eyed female with a red-eyed male. All female offspring had red eyes and all male offspring had white eyes.

Conclusion: The gene for white eye is present on the X chromosome, and the Y chromosome does not have the allele of this gene. Morgan called such genes sex-linked genes. (W = red, w = white)

$F_1 \times F_1$	X^W (sperm)	Y (sperm)
X^W (egg)	X^WX^W red female	X^WY red male
X^w (egg)	X^WX^w red female	X^wY white male

11. Describe sex-influenced and sex-limited traits with common examples from human genetics.

Some traits are not sex-linked (their alleles are not on the sex chromosomes), but their ratio is different in males and females. They are of two categories.

1. Sex-limited traits

The genes of sex-limited traits are present in both sexes but are expressed in only one sex, due to anatomical or hormonal differences. Examples: genes of horn development are expressed only in male sheep; genes of milk production are expressed only in female mammals; in humans, the genes for growth of beard and for prostate cancer are expressed only in males, and the genes for ovarian cancer only in females.

2. Sex-influenced traits

In sex-influenced traits, the same gene shows different expression in each sex. Sex-influenced genes have alleles which are dominant in one sex and recessive in the other, due to the hormonal difference between males and females. Examples in humans: amount of body hair and baldness. The allele of baldness is dominant in men but recessive in women. So a man can be bald when he is homozygous or heterozygous for the baldness allele, while a woman can be bald only when she inherits two baldness alleles.

12. Write a note on the inheritance of haemophilia in humans.

Haemophilia is a rare X-linked recessive trait. In this disease, the blood fails to clot properly after an injury. It is due to the reduction, malfunctioning or absence of blood clotting factors. It is of three types:

- **Haemophilia A** — abnormal clotting factor VIII; X-linked recessive.
- **Haemophilia B** — abnormal clotting factor IX; X-linked recessive.
- **Haemophilia C** — abnormal clotting factor XI; autosomal recessive, so present in both genders equally.

Inheritance of haemophilia A and B: Being X-linked recessive, they are more common in males than females. A male with one recessive allele (X^hY) is haemophiliac. A female is haemophiliac only when she has two recessive alleles (X^hX^h); with one allele (X^HX^h) she is a normal carrier. For example, a carrier mother and a normal father:

Gametes	X^H (sperm)	Y (sperm)
X^H (egg)	X^HX^H normal daughter	X^HY normal son
X^h (egg)	X^HX^h carrier daughter	X^hY haemophiliac son

So 50% of the sons of a carrier mother can be haemophiliac. A haemophiliac father passes the allele to all his daughters (carriers) but to none of his sons.

13. Write a note on the inheritance of colour blindness in humans.

In people with normal (trichromatic) vision, there are three kinds of cone cells in the retina: red-sensitive, green-sensitive and blue-sensitive cones. Each contains its specific pigment protein called **opsin**. The genes for red and green opsins are on the X chromosome, while the gene for blue opsin is on autosome 7. The recessive alleles for opsins cause colour blindness.

- **Dichromacy:** perception of two primary colours and blindness of the third, whose opsin is missing. It may be red-blindness (protanopia), green-blindness (deuteranopia) or blue-blindness (tritanopia).
- **Monochromacy** (total colour blindness): perception of a single colour and blindness of the two colours whose opsins are missing. In blue-cone monochromacy, red and green opsins are absent; it is also called red–green colour blindness.

Inheritance: Red–green colour blindness is an X-linked recessive trait, so it is more common in males. A male needs only one recessive allele on his X chromosome to be colour-blind, while a female needs two. A carrier mother can pass it to 50% of her sons; a colour-blind father makes all his daughters carriers.

There is no cure. Colour-blind glasses, special software and colour discrimination training can help.

14. Explain Morgan’s experiment to explain crossing over and gene linkage.

Morgan’s experiment (gene linkage)

Thomas Hunt Morgan studied two traits in *Drosophila*: wing length and abdomen width. The allele for long wings (L) is dominant over vestigial wings (l), and the allele for broad abdomen (B) is dominant over narrow abdomen (b).

1. He crossed a fly with long wings and broad abdomen with a fly with vestigial wings and narrow abdomen. All F_1 flies had long wings and broad abdomens.
2. He then crossed two F_1 flies.
3. **Expected result:** According to the law of independent assortment, F_2 should show a 9:3:3:1 ratio: 9 long broad and 1 vestigial narrow (parental combinations), 3 long narrow and 3 vestigial broad (new combinations).
4. **Observed result:** Morgan observed only parental combinations in F_2: 3/4 flies with long wings and broad abdomen, and 1/4 with vestigial wings and narrow abdomen.

Conclusion: The genes for wing length and abdomen width are located on the same chromosome (linked), so they did not assort independently.

Crossing over

During the formation of gametes (meiosis-I), homologous chromosomes go through **crossing over**. The chromatids of two homologous chromosomes line up next to one another and exchange sections of DNA; the DNA strands actually break and re-join. After crossing over, the homologous chromosomes separate. The chromatids still have the same genes in the same order, but the alleles have been rearranged. When these recombinant chromatids are distributed to different gametes, a wide variety of gametes is produced. So crossing over leads to the recombination of linked genes and is responsible for genetic variability.

If parental combinations are more frequent and a few new combinations also appear in such a cross, it shows that the linked genes were separated by crossing over (incomplete linkage).

INQUISITIVE QUESTIONS

1. Collect data from the class or the college to see how many individuals have AB blood group and construct a pie chart and histogram for the collected data.

This is an activity. Do it in the following steps:

1. Ask every student of the class his or her ABO blood group and record it in a tally table (A, B, AB, O).
2. Count the students of each group and find the percentage: (number in group ÷ total students) × 100.
3. **Pie chart:** angle of each sector = (number in group ÷ total) × 360°. Draw the sectors with a protractor and label them.
4. **Histogram / bar chart:** put the blood groups on the x-axis and the number of students on the y-axis, and draw one bar for each group.

Worked example (sample data only, not real data):

Blood group	Students	Percentage	Angle in pie chart
A	12	24%	86.4°
B	16	32%	115.2°
AB	4	8%	28.8°
O	18	36%	129.6°
Total	**50**	**100%**	**360°**

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2. Use a dice to calculate how many times out of 100 throws can you get sixes.

The probability of getting a six in one throw = 1/6. So the expected number of sixes in 100 throws = 100 × 1/6 = **16.7, i.e., about 16 or 17 times**. When you actually throw the dice 100 times, the result may be a little more or less than this, because each throw is a chance event. The larger the number of throws, the closer the result comes to 1/6.

3. Derive an idea to get alternatives of blood transfusion. (reference could be made to synthesized plasma and serum).

Transfusion problems arise from the antigens on the donor's RBCs and the antibodies in the plasma. So alternatives should avoid these:

- **Synthesized plasma / plasma expanders** (e.g., saline and other fluids): they have no cells and no antigens, so they can restore blood volume in any patient without matching.
- **Serum or separated blood components:** giving only the needed part (plasma, clotting factors, antibodies) instead of whole blood.
- **Artificial oxygen carriers** (synthetic blood substitutes) that carry oxygen but have no ABO or Rh antigens.
- **RBCs grown in the laboratory** from stem cells of the O-negative type, or the patient's own blood stored before surgery.

This topic is not in the textbook; the answer is based on general knowledge.

4. Justify why a recessive blood group allele of 'i' is more frequent in population.

The allele i is recessive, so it can remain hidden in heterozygous people of blood groups A and B and pass to the next generation. Many different crosses produce it:

- O (ii) × O (ii): all offspring have blood group O (ii).
- $I^Ai \times I^Ai$: 25% offspring are O (ii).
- $I^Bi \times I^Bi$: 25% offspring are O (ii).
- $I^Ai \times I^Bi$: 25% offspring are O (ii).

That is why the allele i is more frequent in the population.

5. Justify blood donation as a service to suffering humanity.

- Blood cannot be made in a factory; it can only come from a donor. So a donor directly saves the life of a patient.
- Patients of accidents, surgery and childbirth, and patients of thalassaemia and haemophilia, need blood regularly.
- Babies with erythroblastosis foetalis need their blood to be replaced with Rh-negative blood.
- Rare groups such as O-negative (universal donor) are always in short supply.
- A healthy donor's body replaces the donated blood, so the donor loses nothing while a patient gets a new life.

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6. Suggest ways to save lives through the knowledge gained in this chapter.

- Match the ABO and Rh blood groups of donor and recipient before every transfusion to prevent agglutination.
- Test the Rh group of mother and father; give Rh-antiserum to an Rh-negative mother to prevent erythroblastosis foetalis.
- Use embryonic screening (ultrasound, chorionic villus sampling, amniocentesis) to detect genetic disorders early.
- Use genetic counselling and carrier testing in families with X-linked disorders such as haemophilia and muscular dystrophy.
- Use probability rules to tell parents the chance of an inherited disorder (e.g., albinism) in their next child.

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